# A Cost-Aware Hybrid LLM Pipeline for Municipal Permit Data Normalization

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**Problem**

# Hybrid Permit Data Normalization

- Noisy and inconsistent permit labels
- Rule-based methods fail on ambiguous cases
- Need semantic understanding for better normalization

## Method

1. Raw Label → Rules- baseline → [Easy ✓] → Output

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2. Raw Label → hybird pipeline→ Evaluation against gold labels [Hard ❗ ] → Navigator AI → Output

Main:

- Rule-based for high-confidence cases
- Navigator AI for ambiguous cases
- Cost-aware: AI used selectively

## Results + Demo

Hybrid pipeline outperforms the rule-only baseline on both evaluation sets
500-record set: accuracy improves from 70.0% to 85.6%
1000-record set: accuracy improves from 68.5% to 94.3%
Largest gains come from ambiguous labels that rules often map to “other”

Demo: example + pipeline + results

**Thank you!**